

Global Semiconductors and Semiconductor Capital Equipment

Global Semis: August 2026 WSTS Tracker - Sales up 23.3% MoM, better than typical (+9.7% MoM) and +158% YoY on continued memory strength



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This weekend the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including August 2026.

Total semiconductor sales grew 158% YoY in August after rising 133% YoY in July. YoY strength was again dominated by memory with sales growing 429% YoY, driving ~\$443B in incremental revenue YTD and with memory pricing alone driving almost 80% of the 119% total industry growth seen since the beginning of the year. Excluding memory, sales were still up strongly at 45% YoY.

Total semiconductor sales MoM were above typical seasonality, up 23.3% versus the historical August average of +9.7% MoM, driven by memory strength. On a MoM basis as-reported, 3 product groups were better than typical historical patterns, with 7 below and 1 inline.

Sales were better than typical MoM for Analog App Specific (3.8% vs. typical -0.1%), Logic (13.7% vs. typical 4.5%), and NAND (41.0% vs. typical 32.0%). Discretes (-2.1% vs. typical -0.2%), Optoelectronics (3.3% vs. typical 5.4%), Sensors & Actuators (-5.5% vs. typical 6.7%), Analog Std Linear (-12.9% vs. typical -3.8%), MPU (1.8% vs. typical 5.0%), MCU (-8.7% vs. typical 0.2%), and DSP (-14.5% vs. typical -5.7%), were below typical MoM patterns. DRAM was inline with typical MoM patterns (34.4% vs. typical 34.7%).

By geography, total sales increased YoY and MoM in all regions.

Total unit shipments were flattish at -0.6% MoM, with ASPs down 24.0% MoM. On a YoY basis, shipments increased by 18.8%.

Application specific ICs sales were up 10.3% MoM, better than typical historical patterns (up 3.5% MoM). 5 end markets were better than typical MoM while 1 was worse than typical. Groups that performed better than historical MoM patterns were: Consumer (4.6% vs. typical 2.4%), Computer & Peripherals (11.6% vs. typical 4.6%), Wireless Communication (13.6% vs. typical 5.6%), Wired Communication (18.0% vs. typical 3.0%) and Multipurpose and Others (-3.3% vs. typical -4.2%).

While Automotive (-6.7% vs. typical 0.6%) was worse than typical.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	2 Oct 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
AMD (Advanced Micro)	O	USD	633.91	650.00	258.5%	USD	4.17	7.55	15.56	151.9	84.0	40.7
ADI (Analog Devices)	O	USD	417.15	465.00	57.6%	USD	7.79	12.85	16.77	53.5	32.5	24.9
AVGO (Broadcom)	O	USD	355.14	575.00	(10.0)%	USD	6.82	11.64	18.63	52.1	30.5	19.1
INTC (Intel)	M	USD	119.33	110.00	204.9%	USD	0.43	1.55	1.81	279.9	77.1	65.8
NVDA (NVIDIA)	O	USD	233.95	400.00	8.9%	USD	4.77	9.20	15.80	49.0	25.4	14.8
NXPI (NXP Semiconductors)	M	USD	243.66	290.00	(8.0)%	USD	11.81	15.02	16.85	20.6	16.2	14.5
QCOM (Qualcomm)	M	USD	184.87	165.00	(5.5)%	USD	12.03	10.51	9.57	15.4	17.6	19.3
TXN (Texas Instruments)	M	USD	293.80	290.00	46.1%	USD	5.45	8.33	9.16	53.9	35.3	32.1
PDFS (PDF Solutions)	O	USD	55.50	65.00	97.3%	USD	0.94	1.35	1.75	58.9	41.2	31.7
KLAC	O	USD	206.89	250.00	66.6%	USD	3.76	5.47	7.05	55.0	37.8	29.3
LRCX (Lam Research)	O	USD	347.49	385.00	121.4%	USD	5.82	9.22	10.97	59.7	37.7	31.7
AMAT (Applied Materials)	O	USD	540.04	700.00	126.5%	USD	9.42	12.76	18.30	57.4	42.3	29.5
TSM (TSMC)	O	USD	472.78	554.00	49.1%	USD	10.28	17.18	22.48	14.6	10.4	7.8
2330.TT (TSMC)	O	TWD	2,500.00	3,300.00	57.7%	TWD	66.25	110.70	144.85	12.0	8.5	6.4
2303.TT (UMC)	U	TWD	161.50	88.00	247.5%	TWD	3.34	4.58	5.56	5.3	4.8	4.5
5347.TT (Vanguard)	M	TWD	187.00	146.00	63.4%	TWD	4.30	5.70	6.59	5.5	5.1	4.8
2454.TT (MediaTek)	O	TWD	4,950.00	5,500.00	254.2%	TWD	66.17	68.52	169.35	74.8	72.2	29.2
3034.TT (Novatek)	M	TWD	537.00	480.00	7.5%	TWD	26.87	30.43	32.54	20.0	17.6	16.5
005930.KS (SEC- Samsung)	O	KRW	276,000	440,000	189.3%	KRW	6,611.53	45,072	74,602	41.7	6.1	3.7
005935.KS (SEC-Pref - Samsung)	O	KRW	201,000	374,000	166.3%	KRW	6,611.53	45,072	74,602	30.4	4.5	2.7
SMSN.LI (Samsung)	O	USD	5,120.00	7,857.00	201.8%	USD	116.15	781.29	1,332.17	44.1	6.6	3.8
000660.KS (SK hynix)	O	KRW	1,841,000	2,700,000	344.7%	KRW	60,341	353,627	554,848	30.5	5.2	3.3
MU (Micron)	O	USD	1,074.89	1,300.00	470.0%	USD	8.29	73.78	173.33	129.7	14.6	6.2
285A.JP (KIOXIA)	U	JPY	19,300	49,000	816.1%	JPY	1,014.00	10,485	14,140	19.0	1.8	1.4
688012.CH (AMEC)	O	CNY	318.55	660.00	37.9%	CNY	2.28	5.30	7.26	139.6	60.2	43.9
688041.CH (Hygon)	O	CNY	226.80	450.00	(31.0)%	CNY	1.10	2.04	3.59	206.2	111.0	63.1
002371.CH (NAURA)	O	CNY	615.97	1,380.00	15.3%	CNY	5.66	10.11	19.92	108.8	61.0	30.9
688072.CH (Piotech)	O	CNY	640.00	1,200.00	125.1%	CNY	3.32	10.18	16.61	192.8	62.8	38.5
6415.TT (Silergy)	O	TWD	425.00	650.00	58.1%	TWD	6.40	9.54	16.69	66.4	44.5	25.5
ASML.NA (ASML)	O	EUR	1,653.00	2,500.00	77.7%	EUR	24.72	38.91	53.56	66.9	42.5	30.9
ASML (ASML)	O	USD	1,867.31	2,859.00	66.3%	USD	27.95	44.01	60.58	59.4	37.7	27.4
6857.JP (Advantest)	O	JPY	38,630	45,800	113.6%	JPY	513.30	948.96	1,088.87	75.3	40.7	35.5
6146.JP (DISCO)	O	JPY	60,290	99,000	(14.3)%	JPY	1,245.79	1,830.28	2,336.94	48.4	32.9	25.8
7741.JP (Hoya)	O	JPY	24,180	35,500	(19.6)%	JPY	744.42	829.63	977.78	32.5	29.1	24.7
4062.JP (Ibiden)	O	JPY	12,100	29,000	392.5%	JPY	214.91	329.25	511.84	56.3	36.8	23.6
6525.JP (Kokusai)	O	JPY	10,050	12,420	82.5%	JPY	128.63	262.29	373.69	78.1	38.3	26.9
6920.JP (Lasertec)	O	JPY	46,130	54,000	98.0%	JPY	863.65	1,185.31	1,348.53	53.4	38.9	34.2
6723.JP (Renesas)	O	JPY	3,575.00	6,300.00	64.7%	JPY	181.61	261.36	302.64	19.7	13.7	11.8
3436.JP (SUMCO)	M	JPY	3,588.00	3,290.00	79.2%	JPY	(33.61)	(41.59)	64.43	(106.8)	(86.3)	55.7
7735.JP (Screen)	M	JPY	15,465	15,000	82.7%	JPY	486.61	574.49	759.55	31.8	26.9	20.4
8035.JP (Tokyo Electron)	O	JPY	12,100	79,300	79.8%	JPY	1,250.88	1,687.17	2,635.81	9.7	7.2	4.6
SOI.FP (Soitec)	O	EUR	170.70	215.00	284.6%	EUR	(6.17)	0.01	3.92	(27.7)	N/M	43.6
XFAB.FP (X-Fab)	O	EUR	7.00	12.90	(13.4)%	USD	0.23	(0.25)	0.29	19.8	(345.2)	23.0
SNDK (SanDisk)	O	USD	1,719.99	3,000.00	1270.5%	USD	70.08	238.36	272.09	24.5	7.2	6.3
SPX			7,722.72									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, KLAC, LRCX, 6920.JP, SOI.FP, SNDK base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

Ticker	Rating	Cur	2 Oct 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
ASIA			1,948.97									
EM			1,807.81									
EDME			1,566.67									
JPL			2,670.65									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, PDFS, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP, SNDK valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, KLAC, LRCX, 6920.JP, SOI.FP, SNDK base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

ADI (Outperform, \$465.00): ADI is executing well with both cyclical and idiosyncratic drivers, and valuation now suggesting better risk-reward.

AMD (Outperform, \$650.00): Expectations remain high, but exposure to AI demand driving both a CPU and GPU story can provide substantial growth.

AVGO (Outperform, 575.00): A strong 2026 AI trajectory seems set to markedly accelerate into 2027 and 2028.

INTC (Market-Perform, \$110.00): Server strength is helping the company get back on their feet, and narrative/headlines may fuel the vibe for now

NVDA (Outperform, \$400.00): The datacenter opportunity is enormous..

NXPI (Market-Perform, \$290.00): The pace and makeup of recovery remains open for debate, though company execution is solid and valuation is increasingly inexpensive.

QCOM (Market-Perform, \$165.00): The diversification story is real but a number of near-term headwinds (smartphone memory dynamics, margins, AAPL etc) need to be navigated first.

TXN (Market-Perform, \$290.00): TXN shares feel fully valued in the current environment.

PDFS (Outperform, \$65.00) A transition to analytics provides for growth and margin expansion amid numerous structural drivers, with eProbe tools representing a new meaningful business opportunity.

AMAT (Outperform, \$700.00): Exposure to key inflections is strong & valuation vs peers is attractive

LRCX (Outperform, \$385.00): The company is benefiting from key inflections (GAA, packaging, HBM, NAND upgrades) and CY26/CY27 commentary seems supportive.

KLAC (Outperform, \$250.00): Amid positive WFE trends KLAC possess structural growth drivers, a strong and durable competitive position, lower China replacement risk, and disciplined capital allocation, warranting premium valuation.

TSMC: We rate TSMC Outperform with PT=NT\$3,300.00.

UMC: We rate UMC Underperform with PT=NT\$88.00.

Vanguard: We rate Vanguard Market-Perform with PT=NT\$146.00.

Novatek: We rate Novatek Market-Perform with PT=NT\$480.00.

MediaTek: We rate MediaTek Outperform with PT=NT\$5,500.00.

Samsung Electronics: We rate Samsung Electronics Outperform with target price of KRW 440,000.

SK hynix: We rate SK hynix Outperform with target price to KRW 2,700,000.

Micron: We rate Micron Outperform with target price of US\$1,300.00.

KIOXIA: We rate KIOXIA Underperform with target price of JPY49,000.

SOI.FP: We rate Soitec Outperform with target price of €215.00.

XFAB.FP: We rate X-FAB Outperform with target price of €12.90

NAURA: We rate Naura Outperform with PT of CNY 1,380.00.

AMEC: We rate AMEC Outperform with PT of CNY 660.00.

Piotech: We rate Piotech Outperform with PT of CNY 1,200.00.

Hygon: We rate Hygon Outperform with PT of CNY 450.00.

Silergy: We rate Silergy Outperform with PT of TWD 650.00.

DISCO: We rate DISCO Outperform with PT of JPY 99,000.

Tokyo Electron: We rate Tokyo Electron Outperform with PT of JPY 79,300.

ASML: We rate ASML Outperform with PT of EUR 2,500.00.

Advantest: We rate Advantest Outperform with PT of JPY 45,800.

Screen: We rate Screen Market-Perform with PT of JPY 15,000.

Kokusai: We rate Kokusai Outperform with PT of JPY 12,420.

Lasertec: We rate Lasertec Outperform with PT of JPY 54,000.

SUMCO: We rate SUMCO Market-Perform with PT of JPY 3,290.00.

Hoya: We rate Hoya Outperform with PT of JPY 35,500.

Ibiden: We rate Ibiden Outperform with PT of JPY 29,000.

Renesas: We rate Renesas Outperform with PT of JPY 6,300.00.

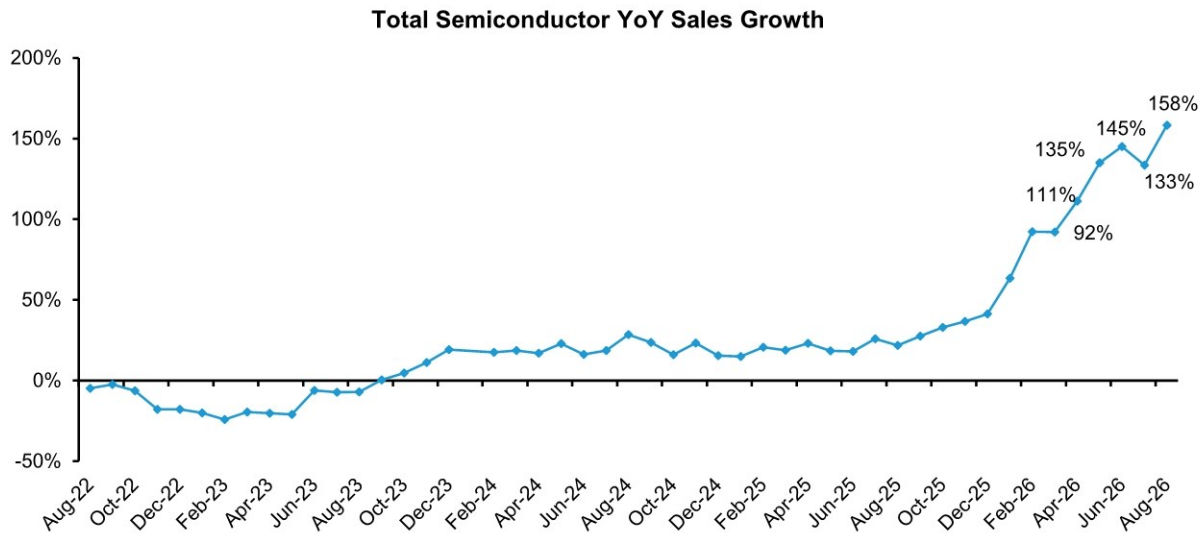
SanDisk: We rate Sandisk Outperform with PT of US\$ 3,000.

DETAILS

This weekend the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including August 2026.

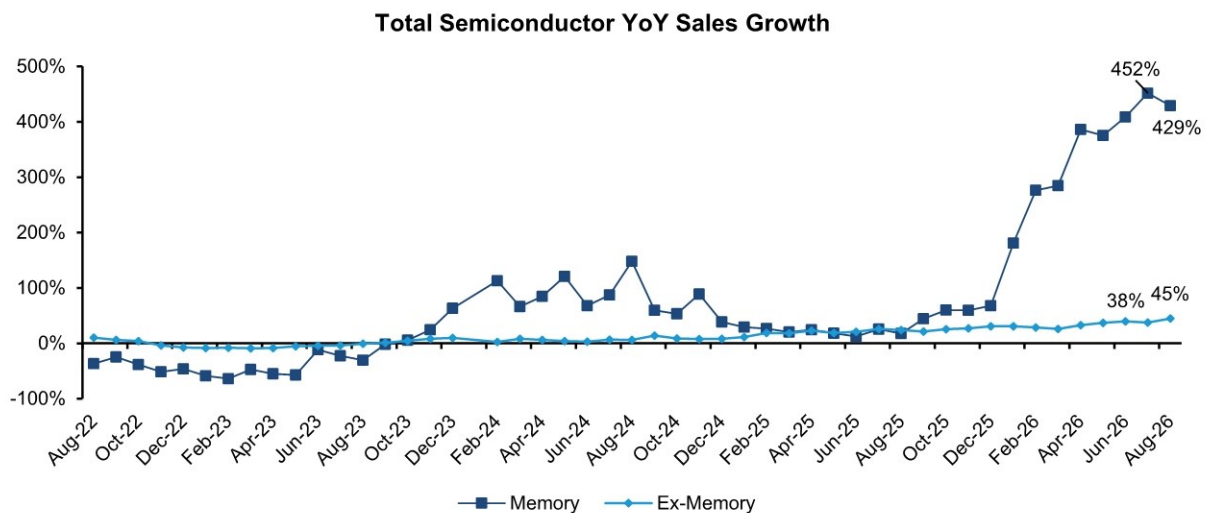
Total semiconductor sales grew 158% YoY in August after rising 133% YoY in July (**Exhibit 1**). YoY strength was again dominated by memory with sales growing 429% YoY, driving ~\$443B in incremental revenue YTD and with memory pricing alone driving ~80% of the 119% total industry growth seen since the beginning of the year (**Exhibit 3, Exhibit 4**). Excluding memory, sales were still up strongly at 45% YoY (**Exhibit 2**). On a 3-month rolling basis sales were up 27.2%, above typical historical patterns of up (4.9%) (**Exhibit 5**).

EXHIBIT 1: **Historical total semiconductor YoY sales growth**

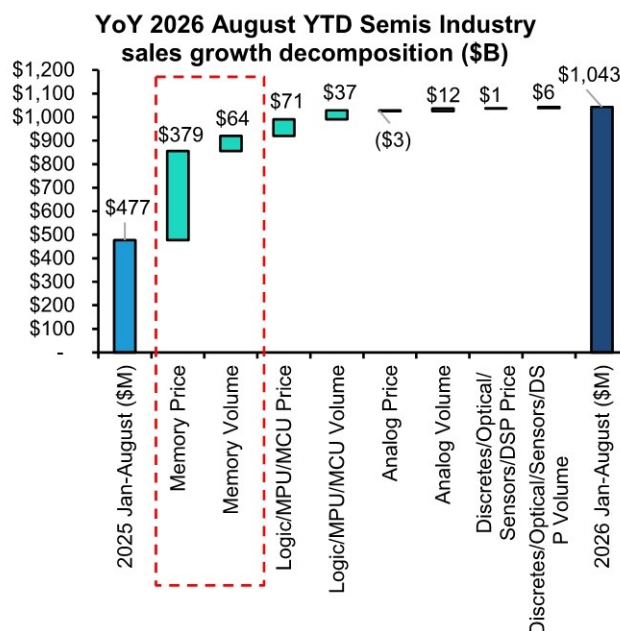


Source: WSTS, Bernstein Analysis

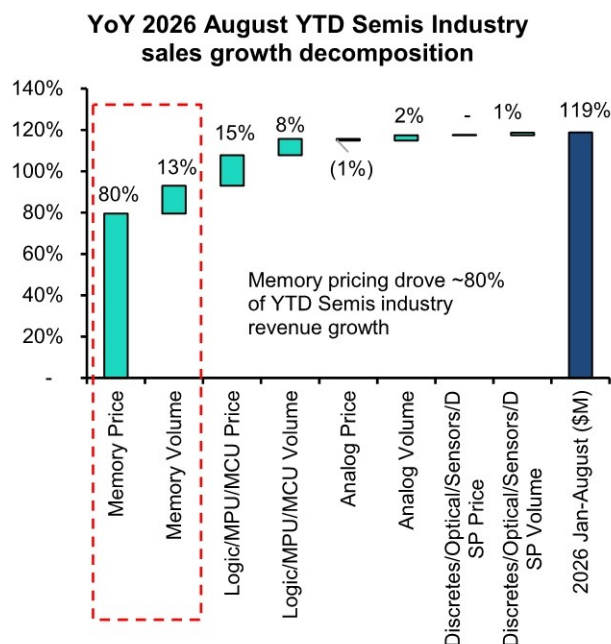
EXHIBIT 2: **Historical memory vs. non-memory YoY sales growth**



Source: WSTS, Bernstein Analysis

EXHIBIT 3: Memory drove ~\$443B in incremental Semis sales YTD through July...


Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 4: ...with Memory pricing alone driving ~80% of the 119% total YTD Semis revenue growth


Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 5: Semiconductor industry % change in sales

	vs 1 month ago		vs 3 months ago		3 months rolling		vs 1 year ago
	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This month
Discretes	-2.1%	-0.2%	8.8%	4.8%	8.0%	3.8%	18.2%
Optoelectronics	3.3%	5.4%	17.1%	18.2%	6.9%	4.8%	9.2%
Sensors & Actuators	-5.5%	6.7%	1.6%	10.3%	3.1%	4.8%	-5.6%
Analog Std Linear	-12.9%	-3.8%	8.4%	7.2%	11.2%	4.2%	15.2%
Analog App Specific	3.8%	-0.1%	20.5%	10.5%	10.0%	5.8%	15.3%
Logic	13.7%	4.5%	24.2%	8.0%	17.8%	6.4%	68.8%
MPU	1.8%	5.0%	15.1%	4.1%	20.9%	2.7%	35.3%
MCU	-8.7%	0.2%	1.2%	7.6%	5.1%	5.1%	15.4%
DSP	-14.5%	-5.7%	5.1%	3.2%	14.5%	1.4%	-4.8%
Memory	36.4%	32.5%	28.8%	3.6%	37.5%	5.7%	429.3%
DRAM	34.4%	34.7%	23.8%	2.9%	33.5%	5.3%	387.7%
NAND	41.0%	32.0%	38.1%	3.5%	45.2%	6.7%	540.8%
TOTAL	23.3%	9.7%	25.0%	6.9%	27.2%	4.9%	158.3%

Source: WSTS, Bernstein Analysis

Total semiconductor sales MoM were above typical seasonality, up 23.3% versus the historical August average of +9.7% MoM. On a MoM basis as-reported, 3 product groups were better than typical historical patterns given memory strength, with 7 below and 1 inline.

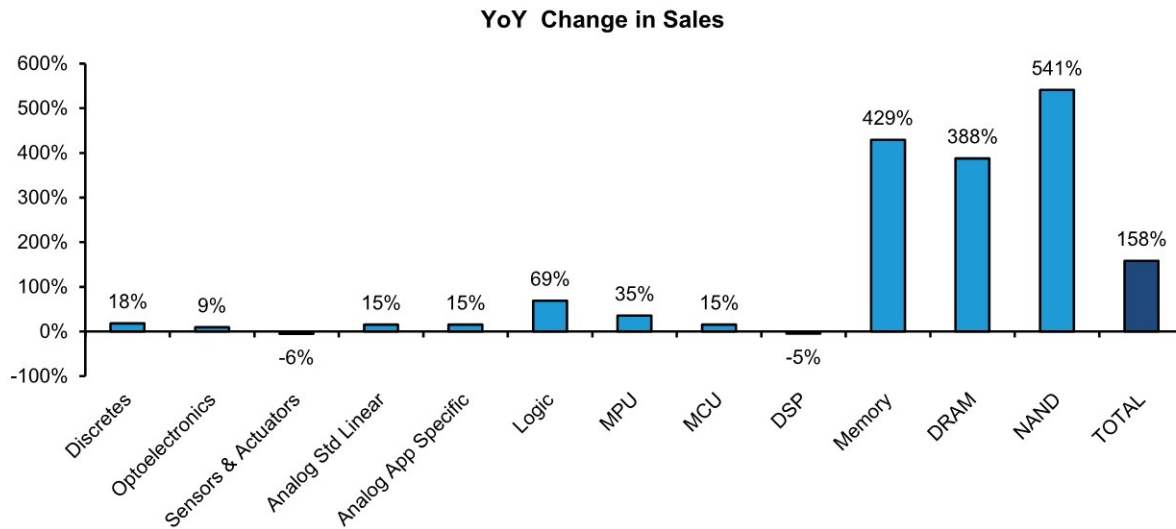
In August on a YoY basis and on a product group level, Discretes, Optoelectronics, Analog Std Linear, Analog App Specific, Logic, MPU, MCU, DRAM, and NAND were up YoY, with Sensors & Actuators and DSP down YoY (**Exhibit 6**).

On a MoM basis, 3 product groups were better than typical historical patterns, with 7 below and 1 inline. (**Exhibit 7**).

- Analog App Specific (3.8% vs. typical -0.1%), Logic (13.7% vs. typical 4.5%), and NAND (41.0% vs. typical 32.0%), were better than typical MoM patterns.

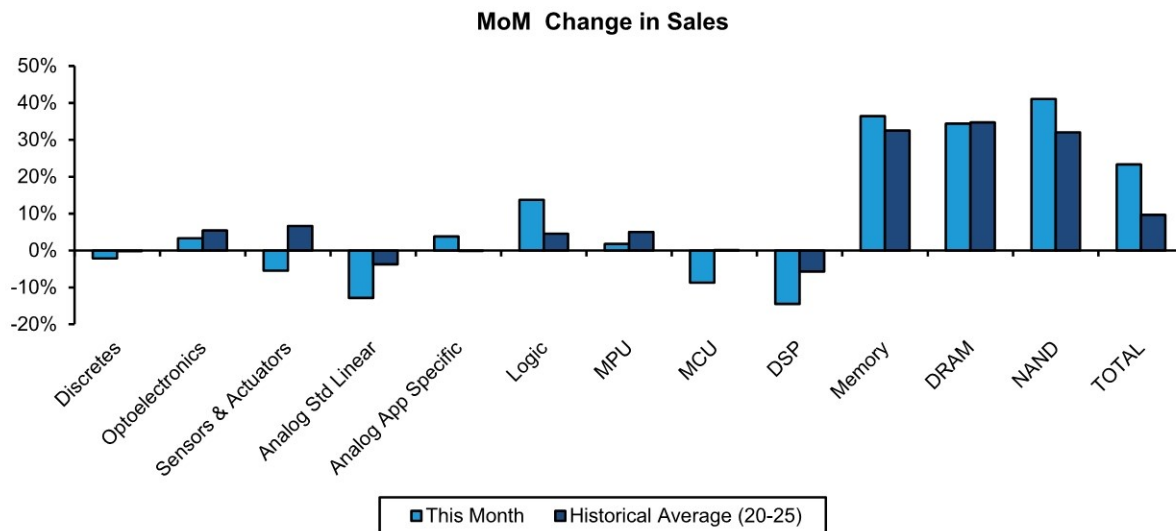
- Discretes (-2.1% vs. typical -0.2%), Optoelectronics (3.3% vs. typical 5.4%), Sensors & Actuators (-5.5% vs. typical 6.7%), Analog Std Linear (-12.9% vs. typical -3.8%), MPU (1.8% vs. typical 5.0%), MCU (-8.7% vs. typical 0.2%), and DSP (-14.5% vs. typical -5.7%), were below typical MoM patterns.
- DRAM was inline with typical MoM patterns (34.4% vs. typical 34.7%).

EXHIBIT 6: Total semiconductor YoY change in sales



Source: WSTS, Bernstein Analysis

EXHIBIT 7: Total semiconductor MoM change in sales



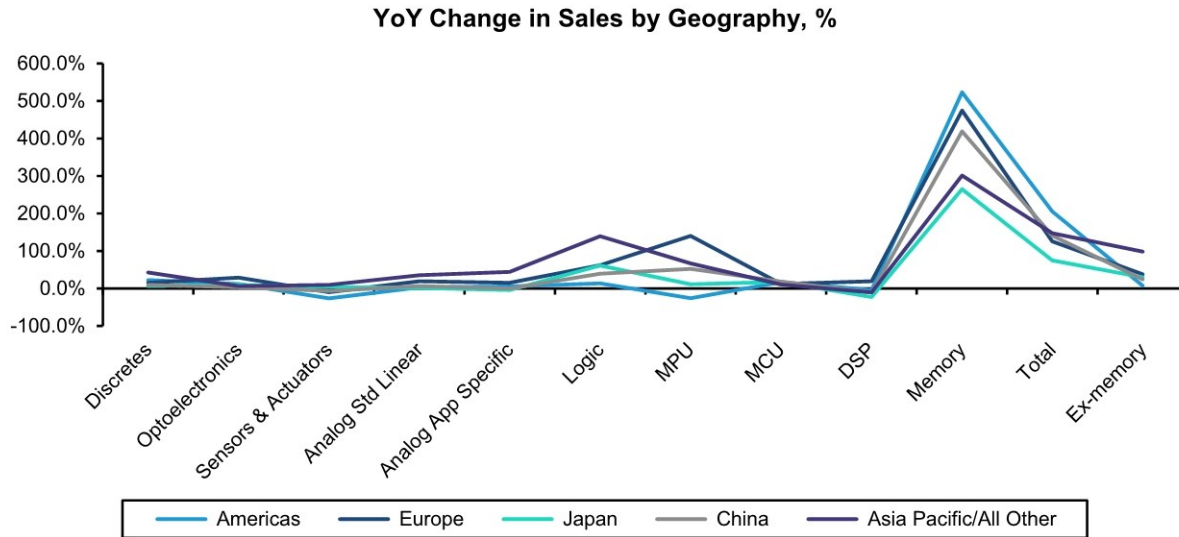
Source: WSTS, Bernstein Analysis

By geography, total sales increased YoY and MoM in all regions.

On a YoY basis, total semiconductors sales were up 206.0% in the Americas, 125.7% in Europe, 75.3% in Japan, 140.9% in China, and 147.3% in Asia Pacific/All Other (**Exhibit 8**). Excluding memory, total sales for these regions were up 7.3%, 38.0%, 30.0%, 24.4%, 98.1%, respectively.

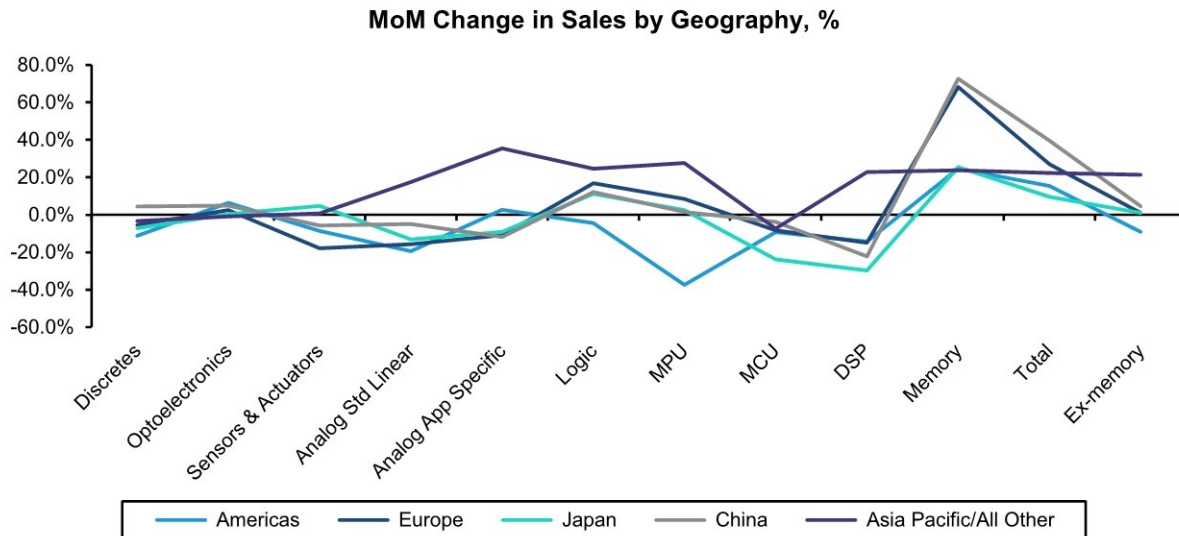
On a MoM basis, sales changes were +15.5% in the Americas, +27.0% in Europe, +9.6% in Japan, +39.6% in China, and +22.4% in Asia Pacific/All Other (**Exhibit 9**). Excluding memory, sales changes for these regions were -9.1%, +1.1%, +1.0%, +4.6%, +21.4%, respectively.

EXHIBIT 8: **Total semiconductor YoY change in sales by geography**



Source: WSTS, Bernstein Analysis

EXHIBIT 9: **Total semiconductor MoM change in sales by geography**



Source: WSTS, Bernstein Analysis

Total unit shipments were flattish at -0.6% MoM, with ASPs down 24.0% MoM.

- 4 out of 11 products groups showed a decrease in unit shipments MoM while 5 groups had MoM unit increases and 2 groups were flattish MoM.
- Units were up MoM for Analog App Specific, Logic, MPU, DRAM[bits], and NAND[bits], (9.8%, 4.5%, 8.1%, 24.8%, 34.5% respectively).
- Units were down MoM for Discretes, Analog Std Linear, MCU, and DSP, (-4.4%, -4.3%, -6.8%, -17.0% respectively).

- Units were flattish MoM for Optoelectronics and Sensors & Actuators, (-0.7%, 0.3% respectively).
- On a 3 months rolling basis overall shipments were up 9.3%.
- On a YoY basis, shipments increased by 18.8%. Units increased for Discretes, Optoelectronics, Analog Std Linear, Analog App Specific, Logic, MPU, MCU, DRAM[bits], NAND[bits], while units decreased for Sensors & Actuators and DSP. (**Exhibit 10**).

EXHIBIT 10: **Change in semiconductor unit shipments**

% Change in Units

	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
Discretes	-4.4%	8.1%	7.1%	7.0%
Optoelectronics	-0.7%	14.0%	13.7%	26.9%
Sensors & Actuators	0.3%	5.9%	6.1%	-3.7%
Analog Std Linear	-4.3%	10.3%	11.6%	30.1%
Analog App Specific	9.8%	14.3%	2.4%	26.4%
Logic	4.5%	12.1%	9.7%	33.7%
MPU	8.1%	23.9%	21.8%	20.7%
MCU	-6.8%	-2.2%	2.6%	4.0%
DSP	-17.0%	3.2%	12.9%	-10.0%
DRAM (Based on Bit Shipped)	24.8%	4.3%	6.0%	27.6%
NAND (Based on Bit Shipped)	34.5%	16.0%	9.4%	29.1%
TOTAL	-0.6%	10.9%	9.3%	18.8%

Source: WSTS, Bernstein Analysis

On a MoM basis, industry ASPs were up 24.0%. ASPs rose in 6 product groups and fell in 5 groups MoM.

- ASPs increased MoM for Discretes, Optoelectronics, Logic, DSP, DRAM[bits], NAND[bits], saw increases (1.5%, 4.1%, 8.8%, 3.0%, 7.7%, 4.8% respectively)
- ASPs decreased for Sensors & Actuators, Analog Std Linear, Analog App Specific, MPU, MCU, (-5.7%, -8.8%, -5.4%, -5.8%, -2.1% respectively).
- Versus 3 months ago, 7 out of 11 product groups saw ASP increases, 1 flattish, and 3 seeing decreases, while on a YoY basis ASPs rose for 7 groups and fell for 4 groups. Total industry ASPs rose 117.4% YoY on the back of memory price increases. (**Exhibit 11**).

EXHIBIT 11: **Change in semiconductor ASPs**

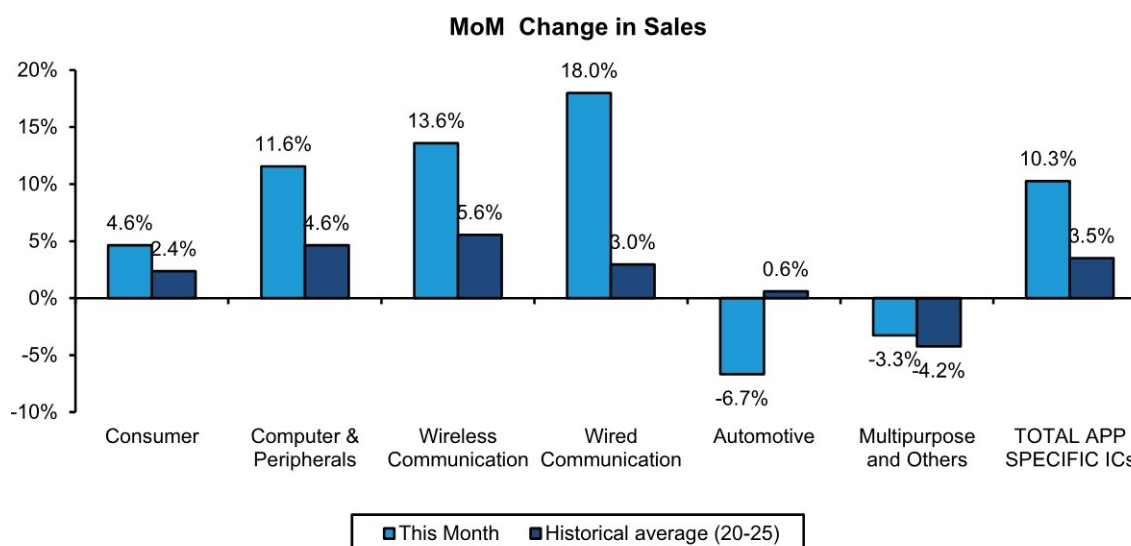
% Change in ASP

	vs 1 month ago	vs 3 months ago	vs 1 year ago
Discretes	1.5%	0.0%	9.8%
Optoelectronics	4.1%	2.8%	-13.8%
Sensors & Actuators	-5.7%	-4.0%	-1.9%
Analog Std Linear	-8.8%	-1.7%	-11.2%
Analog App Specific	-5.4%	5.4%	-8.8%
Logic	8.8%	10.8%	26.3%
MPU	-5.8%	-7.1%	12.1%
MCU	-2.1%	3.3%	10.9%
DSP	3.0%	1.8%	5.7%
DRAM (Based on \$/Bit)	7.7%	18.7%	282.3%
NAND (Based on \$/Bit)	4.8%	19.1%	396.5%
TOTAL	24.0%	12.7%	117.4%

Source: WSTS, Bernstein Analysis

Application specific ICs sales were up 10.3% MoM, better than typical historical patterns (up 3.5% MoM). 5 end markets were better than typical MoM while 1 was worse than typical (Exhibit 12).

- 5 product groups were above typical MoM: Consumer (4.6% vs. typical 2.4%), Computer & Peripherals (11.6% vs. typical 4.6%), Wireless Communication (13.6% vs. typical 5.6%), Wired Communication (18.0% vs. typical 3.0%) and Multipurpose and Others (-3.3% vs. typical -4.2%).
- 1 group was worse than typical: Automotive (-6.7% vs. typical 0.6%).

EXHIBIT 12: **Change in sales by end markets (where WSTS data provides a split by application)**

Source: WSTS, Bernstein Analysis

By product category, we note the following:

Analog Standard Linear: Total Analog Standard Linear sales were down (-12.9%), worse than typical patterns (-3.8%). Shipments were down(-4.3%) with ASPs down (-8.8%). Product category changes MoM were as follows: Amplifiers/Comparators (-23.0% vs. typical -4.1%), Interface (-11.7% vs. typical -0.8%), Power Management (-5.3% MoM vs. typical -3.0% MoM), and Signal Conversion (-30.7% vs. typical -7.7%) (**Exhibit 13**).

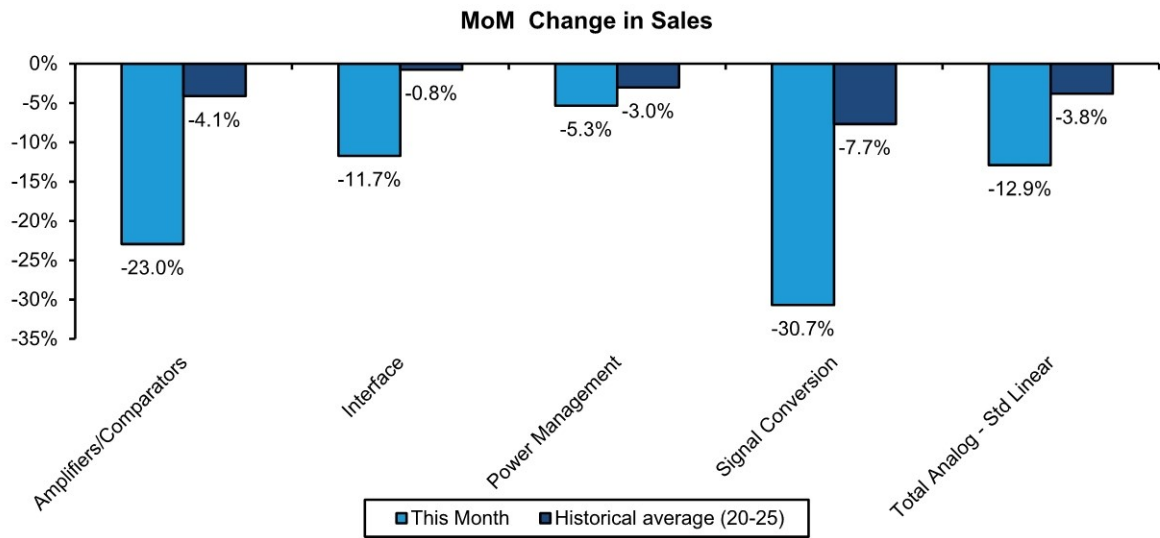
Analog Application Specific: Total analog application specific sales were up (3.8%) MoM, better than typical of -0.1% MoM. Units were up (9.8%) MoM, with ASPs down (-5.4%). End markets changes MoM were as follows: Consumer (12.9% MoM vs. typical 7.2% MoM), Computer (9.0% vs. typical 1.9%), Communications (13.1% MoM vs. typical 1.2% MoM), Automotive (-8.3% MoM vs. typical -1.1% MoM), and Industrial & Others (-1.9% vs. typical of -9.9%) (**Exhibit 14**).

MPU: MPU sales increased 1.8% MoM, worse than typical patterns (5.0%MoM). Units were up (8.1%) MoM with ASPs down (-5.8%) MoM. On a 3-months rolling basis (e.g. June-August vs. March-May), MPU sales increased (20.9%), better than typical average of 2.7%. Sales increased 15.1% vs. 3 months ago and increased 35.3% YoY.

MCU: MCU sales were down (-8.7%) MoM, roughly flat historical averages of roughly flat 0.2%, with units down (-6.8%) and ASP down (-2.1%). Consumer was roughly flat -0.1% MoM vs. typical 4.4% MoM, Computer was -8.3% MoM vs. typical 4.9% MoM, Handsets were down -27.5% MoM vs. typical of up 10.4% MoM, Wireless Communication-All Other was down -10.0% MoM vs typical -3.6%, Wired Communication was up 14.5% MoM vs. typical of 11.1% MoM, Automotive was down -14.8% MoM vs. typical 2.1% MoM, and Multi-purpose and others were -5.2% MoM vs. typical of -3.3% MoM (**Exhibit 15**).

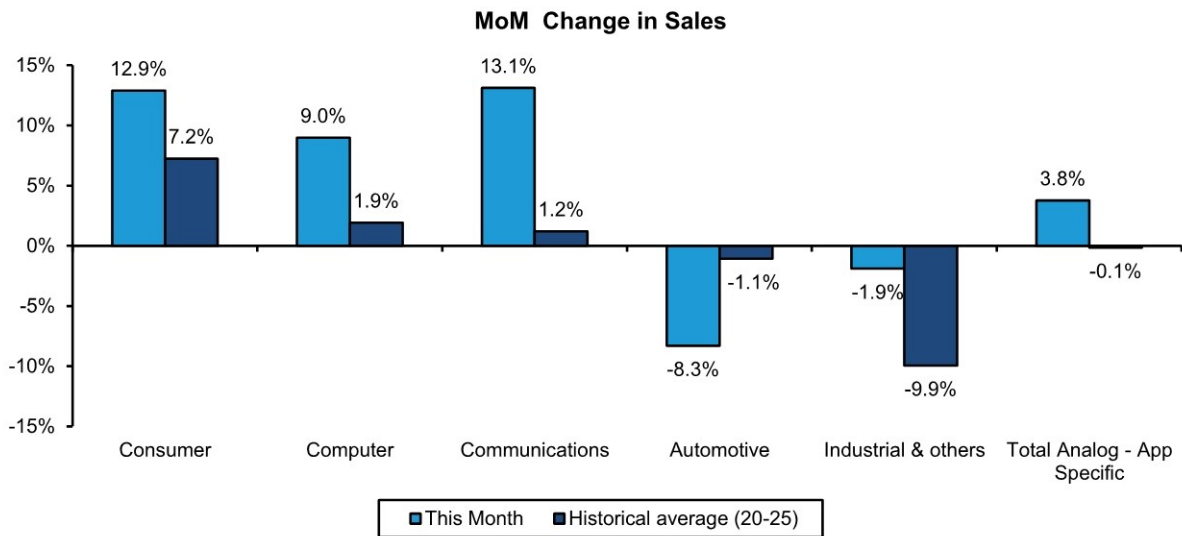
Memory: Memory sales were up (36.4%) MoM, better than historical average of up 32.5%. On a bit shipped basis, DRAM shipments were up 24.8% MoM and NAND up 34.5% MoM. On a \$ per bit basis, DRAM saw ASPs up 7.7%, while NAND ASPs up (4.8%) (**Exhibit 16, Exhibit 17**).

EXHIBIT 13: Analog Standard Linear - MoM change in sales



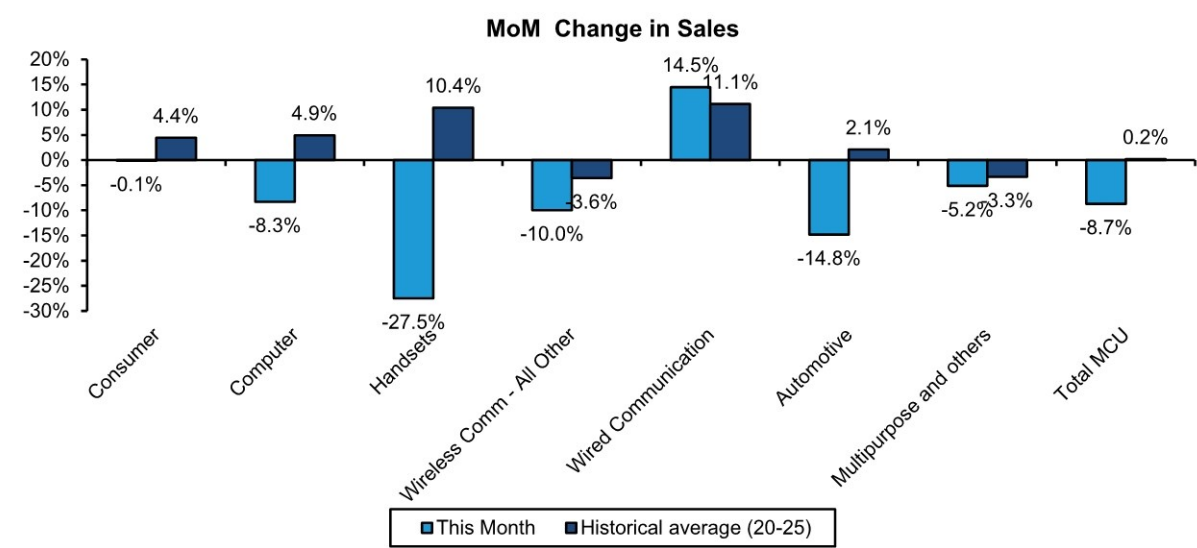
Source: WSTS, Bernstein Analysis

EXHIBIT 14: Analog App Specific - MoM change in sales



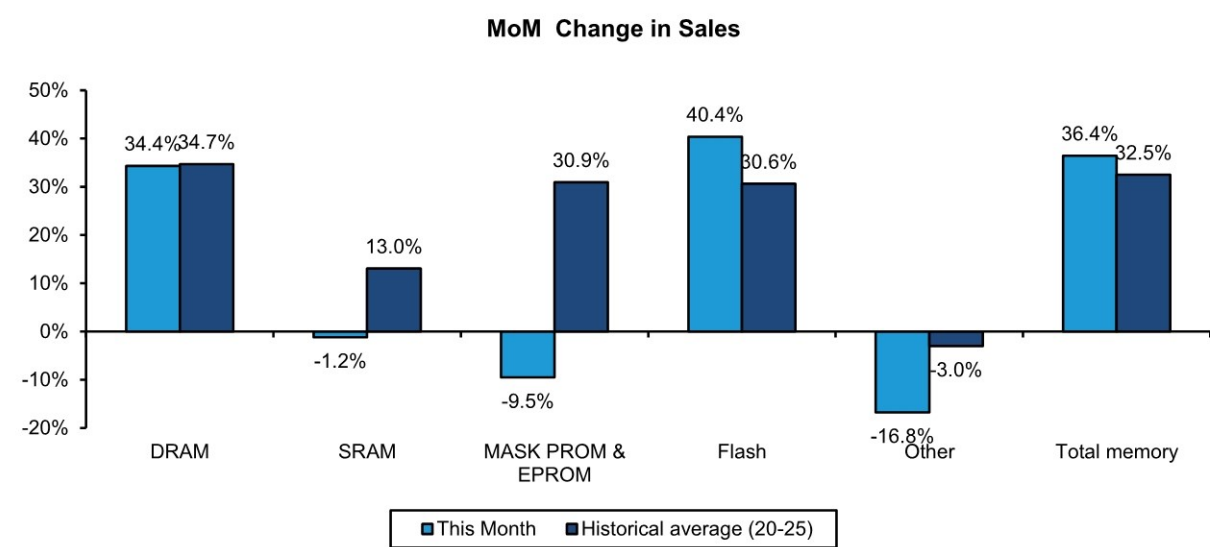
Source: WSTS, Bernstein Analysis

EXHIBIT 15: MCU - MoM change in sales



Source: WSTS, Bernstein Analysis

EXHIBIT 16: Memory - MoM change in sales



Source: WSTS, Bernstein Analysis

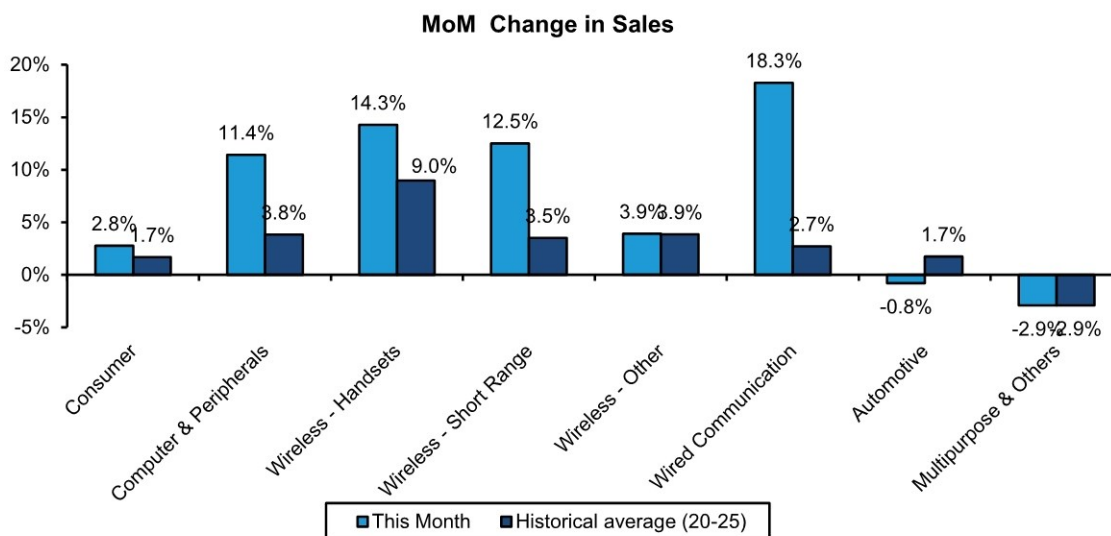
EXHIBIT 17: Memory - Change in units and ASP on bit basis
% change based on bit shipped

	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
DRAM	24.8%	4.3%	6.0%	27.6%
NAND	34.5%	16.0%	9.4%	29.1%

% change in \$ per bit

	vs 1 month ago	vs 3 months ago	vs 1 year ago
DRAM	7.7%	18.7%	282.3%
NAND	4.8%	19.1%	396.5%

Source: WSTS, Bernstein Analysis

EXHIBIT 18: MOS Special Purpose Logic - MoM change in sales


Source: WSTS, Bernstein Analysis

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